

Cloud Computing and AWS

1. Introduction to Cloud Computing

Cloud Computing is a technology that provides computing resources such as **servers, storage, databases, networking, software, and processing power over the Internet.**

Instead of purchasing and maintaining physical computer systems, users can access these resources from a cloud service provider whenever required.

Example

Suppose a company needs a server to host its website.

In the traditional approach, the company needs to:

- Purchase a physical server.
- Install an operating system.
- Maintain hardware.
- Provide electricity and cooling.
- Manage backups and security.

With cloud computing, the company can rent a virtual server from a cloud provider and use it through the Internet.

Definition

Cloud Computing is the delivery of computing resources and services over the Internet on demand.

2. What is AWS?

AWS (Amazon Web Services) is a cloud computing platform provided by **Amazon.**

AWS provides a large collection of cloud services that can be used to build, deploy, store, and manage applications and data without requiring organizations to purchase and maintain all the physical infrastructure themselves.

Major AWS Services

AWS Service	Purpose
Amazon EC2	Provides virtual servers for running applications
Amazon S3	Stores files, images, videos, backups, and other data
Amazon RDS	Provides managed relational databases
AWS Lambda	Runs code without managing servers
Amazon VPC	Creates and manages virtual networks
Amazon Route 53	Provides DNS and domain-related services

Amazon CloudFront	Delivers content quickly to users
Amazon EKS	Provides managed Kubernetes services

Example

Suppose a developer creates an online shopping website.

The developer can use:

- **EC2** → To run the web application
- **S3** → To store product images
- **RDS** → To store customer and product information
- **VPC** → To create a secure network
- **Route 53** → To manage the website domain

Therefore, AWS provides different services required to develop and operate an application.

3. Why Do We Use AWS?

AWS is widely used because it provides flexible and scalable cloud infrastructure.

1. Cost Saving

Organizations do not always need to purchase expensive physical servers. They can use cloud resources according to their requirements.

2. Scalability

AWS allows organizations to increase or decrease computing resources according to demand.

Example:

An e-commerce website may require more servers during a festival sale and fewer resources during normal days.

3. Pay-as-you-go

Many AWS services follow a usage-based pricing model. Users generally pay for the resources they consume instead of purchasing all infrastructure in advance.

4. Global Availability

AWS provides cloud infrastructure in different geographical regions, allowing organizations to deploy applications closer to their users.

5. High Availability

AWS provides services and features that help organizations design applications with redundancy and improved availability.

6. Security

AWS provides various security services and features for controlling access, protecting data, monitoring resources, and securing applications.

7. Wide Range of Services

AWS provides services for:

- Computing
- Storage
- Databases
- Networking
- Artificial Intelligence
- Machine Learning
- Security
- Analytics
- DevOps
- Containers
- Serverless computing

4. Cloud Service Models

Cloud services are commonly divided into three major service models:

1. **IaaS – Infrastructure as a Service**
2. **PaaS – Platform as a Service**
3. **SaaS – Software as a Service**

These models differ according to **how much the cloud provider manages and how much the customer manages.**

5. IaaS – Infrastructure as a Service

IaaS stands for Infrastructure as a Service.

IaaS provides basic computing infrastructure through the Internet.

It can include:

- Virtual machines
- Storage
- Networking
- Virtual networks
- Computing resources

The customer generally manages the operating system, applications, and data, while the cloud provider manages the underlying physical infrastructure.

Example

Amazon EC2 is an example of an AWS service commonly associated with IaaS.

A user can create an EC2 virtual machine and install:

- Windows or Linux
- Required software
- Web server
- Application
- Other tools

Real-Life Example

Consider renting an empty office.

The building and basic facilities are provided to you, but you are responsible for arranging and managing the things needed for your work.

Similarly:

IaaS provides the basic infrastructure, while the customer manages much of the software environment.

Advantages of IaaS

- High flexibility
- Greater control over infrastructure
- Easy scalability
- No need to purchase physical servers
- Suitable for system administrators and IT teams

6. PaaS – Platform as a Service

PaaS stands for Platform as a Service.

PaaS provides a ready-to-use platform for developing, testing, and deploying applications.

The cloud provider manages much of the underlying infrastructure, including:

- Servers
- Operating system
- Networking
- Runtime environment
- Platform infrastructure

The developer mainly focuses on:

- Application code

- Application configuration
- Data

Example

A developer wants to deploy a web application but does not want to spend time managing servers and operating systems.

A PaaS platform provides the environment required to deploy the application.

Real-Life Example

Consider a fully equipped office where:

- Building is ready
- Electricity is available
- Internet is available
- Furniture is available

You only need to bring your work and start working.

Similarly:

PaaS provides a ready platform so developers can focus mainly on application development.

Advantages of PaaS

- Faster application development
- Less infrastructure management
- Easy deployment
- Useful for developers
- Reduces system administration work

7. SaaS – Software as a Service

SaaS stands for Software as a Service.

SaaS provides ready-to-use software applications over the Internet.

The user does not need to install or manage the underlying infrastructure.

Examples

- Gmail
- Google Docs
- Microsoft 365
- Salesforce
- Zoom

Example

When a user uses Gmail, the user does not need to manage:

- Servers
- Operating systems
- Databases
- Application infrastructure
- Hardware

The user simply opens Gmail, logs in, and uses the email service.

Real-Life Example

Consider eating at a restaurant.

You do not need to:

- Build the restaurant
- Purchase kitchen equipment
- Hire the staff
- Prepare the food

You simply use the finished service.

Similarly:

SaaS provides ready-to-use software to the end user.

Advantages of SaaS

- Easy to use
- No installation in many cases
- No hardware management
- Automatic updates are typically handled by the provider
- Accessible through the Internet

8. Difference Between IaaS, PaaS and SaaS

Feature	IaaS	PaaS	SaaS
Full Form	Infrastructure as a Service	Platform as a Service	Software as a Service
Provides	Infrastructure	Development Platform	Ready-to-use Software
Main Users	IT Administrators	Developers	End Users
Control	High	Medium	Lower
Server Management	Customer manages OS and above	Mostly provider managed	Provider managed

Application Development	Customer's responsibility	Customer's responsibility	Already provided
Example	Amazon EC2	AWS Elastic Beanstalk	Gmail

Easy Way to Remember

IaaS → Infrastructure

PaaS → Platform

SaaS → Software

9. Cloud Deployment Models

Cloud deployment models describe **how cloud infrastructure is owned, deployed, and made available to users**.

The major deployment models are:

1. **Public Cloud**
2. **Private Cloud**
3. **Hybrid Cloud**

10. Public Cloud

A **Public Cloud** is a cloud environment in which computing resources are provided by a cloud service provider and made available to multiple customers.

The physical infrastructure is owned and managed by the cloud provider.

Examples

- AWS
- Microsoft Azure
- Google Cloud

Example

A company wants to host its website but does not want to purchase physical servers.

It can rent computing resources from AWS and deploy its website there.

Characteristics of Public Cloud

- Infrastructure is managed by a cloud provider.
- Multiple customers can use the provider's infrastructure.
- Resources can be scaled according to requirements.
- Customers generally pay for the resources they use.
- It reduces the need to maintain physical infrastructure.

11. Private Cloud

A **Private Cloud** is a cloud environment dedicated to a **single organization**.

The infrastructure is not shared with other organizations as a common cloud environment.

A private cloud may be operated by the organization itself or by a third-party provider on its behalf.

Example

A large organization creates a dedicated cloud environment for its internal departments and applications.

Only that organization uses the environment.

Characteristics of Private Cloud

- Dedicated to one organization
- Greater control over the environment
- Can provide greater customization
- Useful when an organization has specific security, compliance, or infrastructure requirements
- Can require more management and investment than using a public cloud

12. Hybrid Cloud

Hybrid Cloud is a combination of **public cloud and private cloud environments**, connected so that workloads or data can be managed across both environments.

Example

A bank may keep certain sensitive systems in its private cloud while using a public cloud such as AWS for other applications or additional computing resources.

Therefore:

Hybrid Cloud = Private Cloud + Public Cloud

Characteristics of Hybrid Cloud

- Combines public and private cloud environments
- Allows organizations to keep selected workloads in a private environment
- Provides access to public cloud resources when required
- Can provide flexibility and scalability
- Requires proper integration and network connectivity between environments

13. Difference Between Public, Private and Hybrid Cloud

Feature	Public Cloud	Private Cloud	Hybrid Cloud
Infrastructure	Provider's infrastructure	Dedicated to one organization	Combination of both
Users	Multiple customers	Single organization	Organization uses both environments
Control	Comparatively lower	Greater control	Combination
Scalability	High	Depends on infrastructure	High
Cost	Generally lower initial infrastructure investment	Can be more expensive to build/manage	Depends on implementation
Example	AWS	Organization's dedicated cloud	Private infrastructure + AWS

14. Service Model vs Deployment Model

Students often confuse these two concepts.

Service Models

They tell us **what type of service the cloud provider gives us**.

IaaS
PaaS
SaaS

Deployment Models

They tell us **how the cloud infrastructure is deployed and used**.

Public Cloud
Private Cloud
Hybrid Cloud

Important Point

IaaS, PaaS and SaaS are not alternatives to **Public, Private and Hybrid Cloud**.

They describe **different aspects of cloud computing**.

For example, an organization can use a **public cloud** and consume an **IaaS service** from it.

AWS Regions, Availability Zones and Data Centers

These three terms are very important in AWS because they describe **where AWS physically runs its cloud infrastructure**.

What is an AWS Data Center?

A **Data Center** is a physical facility that contains the infrastructure required to run cloud services.

A data center contains things such as:

- Physical servers
- Storage systems
- Networking equipment
- Power systems
- Cooling systems
- Security systems

Simple Example

Think of a **large building filled with thousands of computers and networking devices**. This building is a data center.

When you create an EC2 instance on AWS, your virtual machine ultimately runs on **physical hardware inside an AWS data center**.

Data Center = Physical facility containing computing infrastructure.

2. What is an AWS Region?

An **AWS Region** is a **geographical area of the world that contains multiple Availability Zones**.

AWS has Regions in different parts of the world.

For example:

- **Asia Pacific (Mumbai)** → `ap-south-1`
- **Asia Pacific (Singapore)** → `ap-southeast-1`
- **US East (N. Virginia)** → `us-east-1`
- **Europe (Ireland)** → `eu-west-1`

Example

If you select:

Mumbai Region

you are choosing AWS infrastructure located in the Mumbai geographical area.

Region = A geographical location containing multiple Availability Zones.

3. What is an Availability Zone?

An **Availability Zone (AZ)** is an isolated location within an AWS Region.

A Region contains **multiple Availability Zones**.

For example, conceptually:

```
AWS Region
|
|--- Availability Zone 1
|
|--- Availability Zone 2
|
|--- Availability Zone 3
```

Each Availability Zone consists of **one or more data centers**.

So:

Availability Zone = One or more isolated data centers within an AWS Region.

4. Relationship Between Region, Availability Zone and Data Center

This is the most important concept:

```

graph TD
    AWS_Region[AWS Region]
    AZ_A[Availability Zone A]
    AZ_B[Availability Zone B]
    DC_A[Data Centers]
    DC_B[Data Centers]
    PS_A[Physical Servers]
    PS_B[Physical Servers]

    AWS_Region --> AZ_A
    AWS_Region --> AZ_B
    AZ_A --> DC_A
    AZ_B --> DC_B
    DC_A --> PS_A
    DC_B --> PS_B
    AZ_A -.- AZ_B

```

In simple words:

Region → Availability Zones → Data Centers → Physical Servers

5. Real-Life Example

Imagine a large company operating in **Mumbai**.

It has three separate buildings:

```

graph TD
    MR[Mumbai Region] --> AZ1[AZ-1]
    MR --> AZ2[AZ-2]
    MR --> AZ3[AZ-3]
  
```



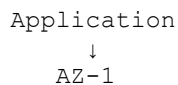
Here:

- **Mumbai** → Region
- **AZ-1, AZ-2, AZ-3** → Availability Zones
- **Buildings containing servers** → Data Centers
- **Servers inside them** → Physical computing infrastructure

6. Why Does AWS Use Multiple Availability Zones?

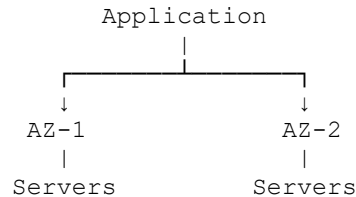
The main reason is to improve **availability and fault tolerance**.

Suppose your application is running only in one location:



If AZ-1 experiences a major failure, your application may become unavailable.

Instead, you can design your application across multiple AZs:



If one Availability Zone has a problem, the application can potentially continue operating from another AZ, depending on how the application is designed.

7. Why Do We Need Multiple Regions?

Organizations may use multiple AWS Regions for reasons such as:

1. Disaster Recovery

If a major problem affects one Region, an organization can maintain resources in another Region as part of its disaster recovery strategy.

2. Lower Latency

A company can deploy applications in a Region geographically closer to its users.

For example, an organization serving users primarily in India may consider an AWS Region in India to reduce network distance.

3. Data Residency

Some organizations have requirements about where their data is stored due to legal, regulatory, or organizational policies.

4. Global Applications

International companies may deploy applications in multiple Regions to serve users around the world.

8. Region vs Availability Zone vs Data Center

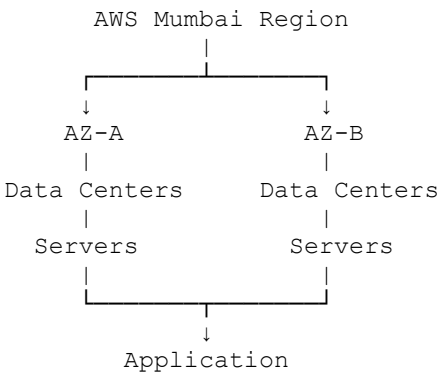
Feature	Region	Availability Zone	Data Center
Meaning	Geographical area	Isolated location within a Region	Physical facility
Contains	Multiple AZs	One or more data centers	Physical IT infrastructure
Level	Highest	Middle	Physical
Example	Mumbai Region	An AZ in Mumbai Region	Physical AWS facility
Main purpose	Geographic deployment	Availability and fault isolation	Running physical infrastructure

9. Important Example for AWS

Suppose you create an application in:

AWS Mumbai Region

You could deploy it across two Availability Zones:



This design can provide better **availability and fault tolerance** than putting everything into a single Availability Zone.